BEU CONNECT

Bihar Engineering University, Patna

End Semester Examination - 2022

Time: 03 Hours

Code: 100304 ***Subject: Data Structure & Algorithms*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

(a) In a stack, if a user tries to remove an element from empty stack it is called:
(i) underflow
(ii) empty collection
(iii) garbage collection
(iv) overflow

(b) Consider the binary max-heap implemented using an array. Which one of the following array represents the heap:
(i) 25, 12, 16, 13, 10, 8, 14
(ii) 25, 12, 16, 13, 10, 8, 14
(iii) 25, 14, 16, 13, 10, 8, 12
(iv) 25, 14, 12, 13, 10, 8, 16

(c) A hash function is defined as h(key) = key mod 7, with linear probing used to insert keys 44, 45, 79, 55, 91, 18, 63 into a table indexed from 0 to 6. What will be the location of key 18?
(i) 3
(ii) 4
(iii) 5
(iv) 6

(d) If the number of values to be sorted is already partially sorted, then _____ sorting can be efficient.
(i) merge
(ii) insertion
(iii) bubble
(iv) selection

(e) The time complexity of merge sort is:
(i) $O(n)$
(ii) $O(\log n)$
(iii) $O(n \log n)$
(iv) $O(n^2)$

(f) State true or false: A: Binary search is used for searching in a sorted array. B: The time complexity of binary search is $O(\log n)$.
(i) True, False
(ii) False, True
(iii) False, False
(iv) True, True

(g) In a circular linked list organization, insertion of a record involves modification of:
(i) One pointer
(ii) Two pointers
(iii) More than two pointers
(iv) No pointer

(h) Level order traversal of a rooted tree can be done by starting from the root and performing:
(i) pre-order traversal
(ii) in-order traversal
(iii) depth first search
(iv) breadth first search

(i) An Abstract Data Type (ADT) is:
(i) same as an abstract class
(ii) a data type that cannot be instantiated
(iii) a data type for which only the operations defined on it can be used, but none else
(iv) all of the above

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

(j) How many distinct BSTs can be constructed with 3 distinct keys?

(i) 4 (iii) 6

(ii) 5 (iv) 7

Q.2 Solve both questions : **[14]**

(a) Explain different asymptotic notations (Big-O, Ω, Θ) used for comparing the time complexity of an algorithm with neat figures.

(b) The run time of an algorithm is represented by the recurrence relation $T(n) = 2T(n/2) + n$; $n$ge2 and with boundary condition $T(1) = 0$. What is the time complexity (in terms of Θ notation).

Q.3 Solve both questions : **[14]**

(a) Discuss pre-order, in-order and post-order traversal techniques of binary tree. Write a C function for non-recursive pre-order traversal.

(b) The pre-order traversal sequence of a Binary Search Tree (BST) is 30, 20, 10, 15, 25, 23, 39, 35, 42. Write step by step process to derive the BST and find post-order traversal also.

Q.4 Solve both questions : **[14]**

(a) Consider a circular queue of capacity n-elements implemented with an array. Write C functions for insertion and deletion operations.

(b) Convert the given infix expression into postfix using stack: A + *B* / C * (*D* + *E*) − F. For each input symbol clearly mention the action taken and status of the stack during conversion.

Q.5 Solve both questions : **[14]**

(a) Write a C function to delete last node from a singly linked list.

(b) Create a max-heap by inserting following keys in the given order. Show each insertion step with clear illustration: 25, 35, 18, 9, 46, 70, 48.

Q.6 Solve both questions : **[14]**

(a) Write an algorithm for merge sort and discuss space and time complexity.

(b) Define collision in hashing. Explain briefly different methodologies to resolve collision.

Q.7 Solve both questions : **[14]**

(a) Write algorithm to count leaf nodes in binary tree. What is the complexity of your algorithm?

(b) Compare BFS and DFS traversal techniques for graph. Write an algorithm to perform BFS using queue.

Q.8 Solve both questions : **[14]**

(a) Differentiate between system defined data types and abstract data types with suitable examples.

(b) What is doubly linked list? What are its applications? Explain how a node can be added as last node using appropriate pseudo code.

Q.9 Write short notes on any two of the following: **[14]**

(a) AVL Rotations

(b) Open Addressing & Chaining

(c) B-Tree

(d) Priority Queue

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store